Response to TCE-2026-04-1669

AURORA: A Multi-Objective Reinforcement Learning Framework for Adaptive Cognitive State Optimization in Virtual Reality

Istiak Ahmed, Ripan Kumar Kundu,
and Khaza Anuarul Hoque, *Senior Member, IEEE*

This report is in response to the comments obtained from the reviewers. According to the comments from the reviewers, we have revised the manuscript and have addressed all of the issues that the reviewers raised to the best of our ability (highlighted in blue in the revised manuscript). We would like to thank the reviewers for their constructive comments, which helped us significantly improve the quality of the paper.

I. Responses to the Comments from the Editor

Concern 1: Scope alignment with TCE. Regarding the scope issue, the manuscript can align to the scope of TCE by incorporating more directives with respect to Consumer Electronics/Technologies. In particular, more CE/CT-related elaborations, illustrations, and citations can be supplemented to highlight the relevance of the contributions to Consumer Technologies/Electronics.

Response:

II. Responses to the Comments from Reviewer-1

Concern 1: Experimental order effects and lack of counterbalancing. The most important concern is the experimental design, particularly the apparent fixed order of the main experimental conditions. According to the current description, the experiment was conducted sequentially as baseline first, static visual techniques next, AURORA thereafter, and AURORA-HIL finally on a separate day. Although the order of the four static techniques

appears to have been randomized within the static condition, the four main conditions do not appear to have been randomized or counterbalanced across participants. If the order was fixed, the observed improvements in AURORA and AURORA-HIL may not be solely attributable to the proposed framework. This concern is particularly important for AURORA-HIL because it was evaluated with the same participants on a separate day after prior exposure to the VR task and after data from earlier conditions had been used for personalization.

Response: We thank the reviewer for this comment and apologize for the confusion. We would like to clarify that the sequential wording in the Experiment Phase description (Section VI) lists the conditions in a narrative order for readability and does not reflect the actual exposure order used in the study. In practice, the study did employ counterbalancing across participants. The Baseline condition (Condition 1) was presented first to establish each participant's reference measurements, the DFOV/DGB/DOF/RF subconditions within Condition 2 were randomized, and the order of the Static (Condition 2) and AURORA (Condition 3) conditions was counterbalanced across participants to mitigate order, fatigue, and learning effects. AURORA-HIL (Condition 4) was the only condition that could not be counterbalanced, because its personalized policy is fine-tuned offline using the feedback and preference data collected from the earlier conditions, and it was therefore necessarily evaluated last on a separate day. We have revised the Experiment Phase description in Section VI to state explicitly the randomization of the static subconditions, the counterbalancing of the Static and AURORA conditions, and the rationale for evaluating AURORA-HIL last.

Concern 2: Insufficient adaptive and ablation baselines. Another major concern is that the comparison baselines are not strong enough to support the authors' broad claims about the necessity and superiority of multi-objective RL. The manuscript compares AURORA mainly against baseline and static visual techniques. Although the comparison between AURORA and AURORA-HIL provides a preliminary assessment of personalization, the manuscript still lacks systematic ablations and stronger adaptive baselines, such as a rule-based adaptive system, single-objective RL for cybersickness only, MTL prediction without RL adaptation, fixed-intensity adaptive policies, and random or greedy policy baselines. Without these comparisons, it remains unclear whether the observed benefits come specifically from the proposed multi-objective RL formulation, from simple adaptivity,

from continuous intensity adjustment, from HIL personalization, or from repeated task exposure.

Response: We thank the reviewer for this constructive comment, and we have strengthened the manuscript with additional baselines to more clearly isolate the source of the observed improvements. Beyond the Static visual-technique conditions, which provide a non-adaptive, fixed-mitigation baseline with real users (Section VI), and the AURORA versus AURORA-HIL comparison, which isolates the contribution of HIL personalization, we have added a focused ablation and baseline study (Section V), in which AURORA is compared in our domain-randomized simulation, under identical settings, against (i) a single-objective RL agent that optimizes cybersickness only, (ii) a fixed-intensity adaptive policy that selects a technique but does not modulate its intensity, and (iii) a rule-based adaptive controller that maps the MTL-predicted cybersickness severity to a technique and intensity without RL. These three baselines respectively isolate the contribution of the multi-objective reward, of continuous intensity control, and of learned RL adaptation over simple rule-based adaptivity. All are evaluated within the same simulator on the same MTL-predicted user states, this comparison isolates the algorithmic contribution of AURORA. To the best of our knowledge, no prior published system jointly optimizes cybersickness, cognitive load, and working memory through adaptive technique-and-intensity control, and therefore a directly comparable external adaptive method is not available. The three baselines above are accordingly constructed to provide a controlled comparison against single-objective, fixed-intensity, and rule-based alternatives.

Concern 3: HIL personalization requires stronger validation. The human-in-the-loop personalization module is an interesting and potentially important contribution, but the current validation is not fully convincing. The manuscript should explain more clearly how user-specific feedback was used for fine-tuning and how the personalized policy was evaluated independently.

Response: We thank the reviewer for this comment. For fine-tuning, the personalization pipeline (Section III(E)) collects two types of feedback from each user. The first is the in-session reports of the user's cognitive state, which are logged together with the currently active visual technique and its intensity. The second is a post-session structured questionnaire that captures technique-specific comfort, perceived visual disruption, preferred intensity level, and an overall technique ranking. These feedback sources are combined into a user-specific preference vector $\mathbf{z}$ that encodes adjustments to the reward weights α_k in Equation 2, technique-specific intensity bounds, and relative preferences over techniques. Between sessions, $\mathbf{z}$ re-weights the reward and constrains the per-technique intensity ranges, and the general AURORA policy is then fine-tuned offline with PPO to obtain the personalized policy. For independent evaluation, the personalized policy (AURORA-HIL, Condition 4 in Section VI) is assessed in a separate session, while the feedback that constructs $\mathbf{z}$ is drawn only from the earlier sessions. The fine-tuning data and the evaluation data are therefore disjoint, so the reported AURORA-HIL results reflect performance on a new session rather than on the data used for personalization. The added benefit of personalization is quantified by the direct AURORA versus AURORA-HIL comparison on the same participants, reported in Table VI. We have clarified these points in Section III(E) and Section VI of the revised manuscript.

Concern 4: Generalizability beyond the VR maze task is limited. The current evaluation is mainly based on a VR maze navigation task involving walking, coin collection, and digit-span recall. While this is a reasonable experimental paradigm, it is still a relatively narrow scenario. The title and several statements suggest a general framework for adaptive cognitive-state optimization in VR. However, the current evidence supports a more limited conclusion: AURORA is effective in a walking-based VR maze task with concurrent cognitive demands. The authors should either provide additional evidence across different VR scenarios or narrow the scope of the title, abstract, and conclusion.

Response: We thank the reviewer for this comment. We would like to clarify that the AURORA framework, comprising MTL-based cognitive-state prediction, the multi-objective reward formulation (Equation 2), and PPO-based adaptation of general-purpose visual techniques (DFOV, DGB, DOF, and rest frame), is methodologically independent of the specific VR scenario. In its current instantiation, however, the prediction model is trained on maze-walking data, and the RL policy is learned within a maze-based simulation environment, so the empirical evidence is confined to this setting. We explicitly acknowledge this single-environment scope as a limitation and identify validation across broader VR tasks and unseen scenarios as a direction for future work (Section VII).

Concern 5: Cognitive-load measurement requires stronger justification. The authors claim that AURORA balances cognitive physical load and cognitive mental load, but these two

outcomes appear to be assessed using a single adapted NASA-TLX question administered once per minute during VR navigation. While such a simplified real-time rating may be practically motivated, it cannot provide sufficiently validated evidence for the strong claim that the system balances cognitive load. Moreover, repeated in-task questioning may itself affect cognitive load, working memory, and cybersickness. The authors should justify this measurement choice, clearly distinguish it from the standard NASA-TLX, discuss its limitations, and provide additional behavioral or objective indicators to support the cognitive-load findings.

Response: We thank the reviewer for this comment. We used a single adapted NASA-TLX item, rather than the full multi-item instrument, specifically to avoid the burden the reviewer notes: repeating the complete NASA-TLX during navigation would itself add mental and physical demand and interfere with the concurrent digit-span working-memory task. This once-per-interval in-task rating is consistent with state-of-the-art VR multitasking and cognitive-state prediction studies [1–6]. It is not a substitute for the standard NASA-TLX [7], which we collected after the study as a validated overall workload measure. The cognitive-load findings also do not rely on this single rating alone. The MTL prediction model objectively estimates CPL and CML in real time from eye- and head-tracking signals (Section III (B)), independent of self-report, and working memory is measured behaviorally through digit-span recall accuracy. We have revised Section VI to make explicit that the single adapted item is used only for in-task measurement, whereas the standard NASA-TLX is collected after the study.

Concern 6: Statistical analysis and reporting require clarification and revision. The statistical analysis and reporting require further clarification and revision. The use of Shapiro-Wilk tests, Friedman tests, and Wilcoxon signed-rank post hoc comparisons with Holm-Bonferroni correction for cybersickness, cognitive physical load, cognitive mental load, and working memory is generally appropriate for repeated-measures nonparametric data. However, several important issues remain. First, the IPQ presence analysis appears to use Mann-Whitney U tests for pairwise comparisons, although the study seems to involve the same participants across multiple conditions. If this is the case, paired or repeated-measures nonparametric tests, such as Friedman tests followed by Wilcoxon signed-rank tests, would be more appropriate. Alternatively, mixed-effects models could be used. Second, the current analyses do not appear to account for potentially important covariates or

confounds, including prior VR experience, baseline sickness level, gender, session/day effects, task familiarization, and condition order. These factors are particularly important because AURORA-HIL was evaluated on a separate day after data from earlier conditions had been used for personalization. Third, the authors should clarify the scope of Holm-Bonferroni correction. It should be clearly stated whether the reported p-values are raw or adjusted, and whether correction was applied within each outcome or across all pairwise comparisons and outcomes. Fourth, the direction of the reported effect sizes should be defined. For example, the manuscript reports negative effect sizes for some improvements in working memory. The authors should clarify whether the sign reflects the direction of the Wilcoxon statistic or the direction of the experimental effect. Finally, the manuscript should check and correct internal inconsistencies in result reporting. For example, the text states that AURORA achieves the lowest cybersickness score, although AURORA-HIL has a lower mean cybersickness score. Such inconsistencies should be corrected to improve the rigor and interpretability of the findings.

Response: We thank the reviewer for these detailed comments.

First, regarding the IPQ analysis, the reviewer is correct. Because the IPQ scores were collected from the same participants across conditions, a paired test is appropriate. We have replaced the Mann-Whitney U test with a Friedman test followed by Wilcoxon signed-rank post hoc comparisons and updated the corresponding results in Section VI.

Second, regarding covariates and confounds, our study uses a within-subjects repeated-measures design in which each participant experiences all conditions and serves as their own control. Stable individual factors such as prior VR experience, gender, and baseline susceptibility are therefore held constant across conditions and do not confound the within-subject comparisons. Condition order was counterbalanced across participants to reduce order and practice effects (see our response to Concern 1), and a warm-up tutorial preceded measurement to limit task familiarization. The remaining concern is that AURORA-HIL was evaluated on a separate day. Prior work has shown that cybersickness symptoms can persist from hours up to several days after VR exposure (https://doi.org/10.1007/s10055-016-0285-9, https://doi.org/10.3357/ASEM.2394.2009), which makes a clean washout or abatement-induced floor effect unlikely. Moreover, a few participants reported noticeable cybersickness after Conditions 1–3 and continued to feel it in Condition 4 on the following day, but felt better with AURORA-HIL.

Third, regarding the Holm-Bonferroni correction, the p-values reported in Table VI are adjusted rather than raw, and the correction was applied within each outcome separately, across the pairwise comparisons within each of CS, CPL, CML, and WM. We have stated this explicitly in the caption of Table VI and in Section VI.

Fourth, regarding the direction of the effect sizes, the reported rank-biserial r reflects the direction of the Wilcoxon statistic, that is, which condition in the pair received higher ranks. Because working memory is scored so that higher is better, whereas CS, CPL, and CML are scored so that lower is better, an improvement appears as a positive r for CS, CPL, and CML but as a negative r for WM. The negative r values for WM therefore indicate improvement. We have added this sign convention to the caption of Table VI.

Finally, regarding internal consistency, the reviewer is correct that AURORA-HIL has the lower mean cybersickness score ($M = 1.12$) than AURORA ($M = 1.52$), so AURORA-HIL achieves the lowest cybersickness. We have corrected this statement and rechecked the manuscript for similar inconsistencies.

Minor Concern 1: The manuscript contains several grammatical and typographical errors, including repeated phrases, malformed words, and inconsistent terminology, for example, "user cognitivstates in page 7", "while while the remaining user in page 8", ". This is less than 27% of the per-frame budget, This is less than 27% of the per-frame budget, remaining within acceptable limits for SOTA real-time VR settings in page 9". A careful language edit is needed.

Response: We thank the reviewer for this comment. We have carefully proofread the entire manuscript and corrected the listed errors, including the repeated phrases and malformed words on pages 7–9, together with other typographical and terminology inconsistencies, in the revised manuscript.

Minor Concern 2: Some abbreviations should be defined consistently at first use.

Response: We thank the reviewer for this comment. We have ensured that all abbreviations (e.g., CS, CPL, CML, WM, DFOV, DGB, DOF, and RF) are defined at first use and used consistently throughout the revised manuscript.

Minor Concern 3: The visual technique library should be described more systematically, including parameter ranges, intensity normalization, and implementation details.

Response: We thank the reviewer for this comment. We have expanded the description of the visual technique library in Section III (D) to present each technique more systematically, including its parameter ranges, intensity normalization, and implementation details, in the revised manuscript.

Minor Concern 4: The limitations section should more explicitly discuss sample size, participant age range, prior VR experience, task specificity, possible order effects, simplified cognitive-load measurement, and the absence of long-term exposure testing.

Response: We thank the reviewer for this comment. We have expanded Section VII (Limitations and Future Work) to explicitly discuss the sample size, participant age range, prior VR experience, task specificity, possible order effects, the simplified cognitive-load measurement, and the absence of long-term exposure testing in the revised manuscript.

Minor Concern 5: The manuscript should avoid overclaiming that AURORA generally optimizes cognitive state in VR unless additional task scenarios are evaluated.

Response: We thank the reviewer for this comment. Consistent with our response to R1C4, we have clarified the scope of our claims and explicitly acknowledge in Section VII that the present evaluation is confined to a single VR task, identifying validation across additional scenarios as future work in the revised manuscript.

III. Responses to the Comments from Reviewer-2

Concern 1.1: Lack of justification for splitting cognitive load. The manuscript introduces Cognitive Mental Load (CML) and Cognitive Physical Load (CPL) as separate metrics derived from the general cognitive load discussed in the Introduction and Background. Although the Introduction lists these metrics as part of the contribution, the paper does not explain why cognitive load should be divided into these two components. From a psychological standpoint, cognitive load is a single construct with intrinsic, extraneous, and germane components (Sweller, 1988) and cannot be meaningfully separated into "mental" and "physical" load.

Response: We thank the reviewer for this important comment. Under Sweller's cognitive load theory [8], cognitive load is a unitary construct with intrinsic, extraneous, and germane components, and we do not intend CPL and CML as a partition of that construct. Instead, CPL and CML follow the multidimensional workload view operationalized by the NASA-TLX [7], which separates physical demand and mental demand as distinct dimensions. The ground-truth labels for these two states are taken from the VRWalking dataset [1] and follow the same separation used in recent VR cognitive-state prediction work [6]. CPL corresponds to the physical-demand dimension and CML to the mental-demand dimension. We keep the two dimensions separate because, during active VR locomotion, they can be separated and are affected differently by the visual techniques the agent controls. Physical demand arises from compensatory head and body movement, postural control, and locomotion effort, whereas mental demand arises from navigation, spatial reasoning, and the interpretation of degraded or incomplete visual information. For example, field-of-view restriction mainly raises physical demand by forcing additional head movement, whereas blur-based techniques mainly raise mental demand by making visual cues harder to interpret. Combining the two into a single value would hide these opposing effects and deny the RL agent the signal it needs to balance them. Regarding terminology, we retain the "cognitive" qualifier because, in active VR, physical demand is not purely muscular. As we discuss in the Introduction, managing compensatory movement under altered visual input consumes additional cognitive and attentional resources, so it carries a cognitive cost beyond the physical one. We therefore use cognitive physical load for the cognitive cost tied to physical and motor demand and cognitive mental load for the cost tied to mental and perceptual processing.

We have added an explicit definition and justification of CPL and CML in the Introduction and Background (Section III), clarifying that they correspond to the physical-demand and mental-demand dimensions of the NASA-TLX rather than to the components of cognitive load theory, and explaining why separating them is meaningful for adaptive visual-technique selection in VR.

Concern 1.2: Insufficient operationalization of UIX and its relationship with the selected cognitive-state metrics. The manuscript repeatedly claims that AURORA improves the overall user immersive experience (UIX), but UIX is not clearly defined or operationalized. While cybersickness is evidently relevant to comfort and immersion, the paper does not explain how the three selected targets—cybersickness, cognitive load, and working memory—jointly constitute or predict UIX. More specifically, these metrics represent different types of constructs: cybersickness is a discomfort/symptom measure, cognitive

load is a workload-related construct, and working memory is closer to task performance or cognitive capacity. The manuscript treats them as a unified optimization target, but does not provide a clear human-factors or HCI rationale for how improvements in these measures should be interpreted as improvements in UIX. This issue is especially important because the RL framework involves trade-offs among these metrics. For example, a technique may reduce cybersickness while increasing workload or impairing task performance. Without defining UIX and explaining how these trade-offs are resolved from a user-experience perspective, the claim that the framework optimizes "overall UIX" remains under-specified. Recommendation: The authors should substantially revise the conceptual framing of cognitive load and UIX. First, they should justify why cognitive load is decomposed into CML and CPL, and clarify whether CPL is intended to represent cognitive load, physical demand, or a broader workload component. If CPL is derived from the NASA-TLX Physical Demand dimension, it should not be labeled as "cognitive" without theoretical justification. Second, the authors should define UIX explicitly and explain how cybersickness, cognitive load, and working memory jointly contribute to this outcome. The revised manuscript should ground these constructs in established cognitive load theory and HCI/user-experience literature, and clearly distinguish between cognitive states, workload dimensions, task performance, and user-experience outcomes.

Response: We thank the reviewer for this comment. The decomposition of cognitive load into CPL and CML and the use of the "cognitive" qualifier are addressed in our response to Concern 1.1.

We use UIX for the overall quality of the user's immersive experience, that is, how comfortable, present, and capable the user feels during immersion. Cybersickness, cognitive load, and working memory are not UIX itself but its determinants. Cybersickness is a discomfort state that lowers comfort and breaks immersion, CPL and CML are workload dimensions that, when high, leave fewer resources for the task, and working memory is a task-performance measure that supports task engagement. The agent optimizes these states, while UIX itself is measured separately through the IPQ presence score collected after each condition, where AURORA reaches a significantly higher presence than the static techniques ($Z = 2.73$, $U = 532$, $p = 0.027$) and AURORA-HIL is higher still, supporting H2. Since a technique can reduce cybersickness while raising cognitive load or lowering working memory, the multi-objective reward in Equation 2 balances these states rather than improving one at the cost of another, and the presence result confirms that the balance yields a better overall experience.

We have added an explicit definition of UIX in the Introduction and clarified in the Background (Section II) that cybersickness, cognitive load, and working memory are determinants of UIX, while UIX itself is reported through the IPQ presence score.

Concern 2: Ambiguity in RL State Feature Encoding. The RL state vector includes multi-modal sensor features and Transformer-predicted cognitive states. However: The manuscript does not provide a high-level explanation of how raw sensor data are processed, normalized, or combined into the RL state vector. Each feature appears arbitrarily mapped, with no methodological justification in the text. Although the system is open-source and reproducible from code, the manuscript alone does not allow readers to interpret how RL actions relate to cognitive states and rewards. Recommendation: Include a high-level diagram or description illustrating how multi-modal features and Transformer outputs form the RL state.

Response: We thank the reviewer for this helpful comment. The RL formulation is defined in Section III (C) as a Markov decision process, with the state, action, transition, reward, and discount factor specified in turn, and the closed loop between the sensor stream, the prediction model, and the agent is shown in the framework overview (Fig. 1). We expand the state description below and make the data flow explicit in the revised manuscript. The RL state is built only from streaming sensor signals and the previously applied action. It contains locomotion signals (walking speed and angular velocity), head-tracking signals (position, rotation, and rotational velocity), eye-tracking signals (pupil diameter, gaze direction, and fixation stability), a visual motion cue (optic flow magnitude), and task context (turning versus straight walking), together with the last visual technique and its intensity. The Transformer-predicted cognitive states (CS, CPL, CML, and WM) are not part of the state. They are mapped by Equation 1 to normalized scores in $[0, 1]$ and enter the reward in Equation 2. The state describes the user's current behavior, while the predicted states describe the user's response and are used only to compute the reward. Each raw signal is read from the VR runtime over a short temporal window, scaled to a common normalized range, and combined into a single fixed-length state vector, so that no single feature dominates simply because of its raw scale. The feature groups are also not arbitrary. Therefore, head motion and optic flow are established drivers of cybersickness, pupil diameter, gaze, and fixation stability track cognitive and perceptual demand, and locomotion and task context capture the physical effort of active walking. Each group is therefore informative for at least one of the targeted states the agent must balance. This also defines how actions relate to cognitive states and rewards. At each step the agent selects a visual technique and intensity, the rendering change alters the user's motion and gaze behavior, and these changes appear in the next state. The prediction model reads the same stream and estimates the four cognitive states, and the change in those estimates between consecutive steps forms the reward in Equation 2. The agent thus learns the link between its actions and their effect on user state through the reward rather than through any hand-set mapping.

We have expanded Section III (C) with a high-level description of how the raw streaming signals are windowed, normalized, and combined into the RL state vector, and we have clarified that the Transformer outputs shape the reward rather than the state, with the overall data flow shown in the framework overview (Fig. 1).

Concern 3: Clarification Needed on the User Study Design and Repeated-Measures Structure. The user study appears to be a within-subjects repeated-measures design, since the manuscript states that participants completed all study conditions, including the baseline, static techniques, AURORA, and AURORA-HIL. The use of Friedman tests and Wilcoxon signed-rank tests also implies a paired/repeated-measures structure. However, the current description is somewhat unclear. The paper first describes four broad experimental conditions: baseline, static visual techniques, AURORA, and AURORA-HIL. Yet in the results, the four static techniques—DFOV, DGB, DOF, and RF—are analyzed separately, resulting in seven analyzed levels: Baseline, DFOV, DGB, DOF, RF, AURORA, and AURORA-HIL. This ambiguity should be clarified because the interpretation of the statistical tests depends on the exact experimental structure. In particular, the authors should specify: whether all 47 participants experienced all seven analyzed conditions; whether DFOV, DGB, DOF, and RF were treated as separate repeated-measures conditions or as subconditions within a single "static techniques" condition; how the order of conditions was randomized or counterbalanced; whether potential order effects, fatigue effects, learning effects, or carryover effects were controlled, especially given that AURORA-HIL was conducted on a separate day with the same participants. This is especially important because VR sickness, task familiarity, and working memory performance can all be sensitive to exposure order and accumulated fatigue. The study design is potentially valid, but the manuscript should

describe the repeated-measures structure and counterbalancing procedure more explicitly.

Response: We thank the reviewer for this comment and confirm that the study is a within-subjects repeated-measures design in which all 47 participants experienced every condition. To clarify the structure: the experiment comprises four conditions (Baseline, Static, AURORA, and AURORA-HIL). Within the Static condition, the four visual techniques (DFOV, DGB, DOF, RF) are administered as subconditions, each applied individually at a fixed intensity. In the results, these four static subconditions are reported separately so that each can be compared directly against Baseline and AURORA, which yields the seven analyzed levels (Baseline, DFOV, DGB, DOF, RF, AURORA, AURORA-HIL). Because the same participants are measured across all levels, the Friedman and Wilcoxon signed-rank tests are applied as paired/repeated-measures procedures. Counterbalancing and the control of order, fatigue, learning, and carryover effects are addressed in detail in our response to Reviewer-1 (Concern 1). We have revised Section VI to make the repeated-measures structure and the seven analyzed levels explicit.

IV. Responses to the Comments from Reviewer-3

Concern 1: The paper claims to be the "first" work to model the optimization of a VR user's cognitive state as a multi-objective RL problem. However, AURORA's primary contribution lies in jointly predicting multiple cognitive states and utilizing them for reward shaping; yet, this fundamentally constitutes a combination of multi-task prediction and RL, lacking any groundbreaking theoretical innovation.

Response: We thank the reviewer for this comment. AURORA is positioned as an applied contribution for consumer VR rather than as a new RL algorithm, so its novelty lies in the problem formulation, the working real-time system, and the empirical findings rather than in a new learning-theoretic result. To the best of our knowledge, AURORA is the first to formulate VR cognitive-state management as a multi-objective RL problem that jointly optimizes cybersickness, cognitive physical load, cognitive mental load, and working memory through adaptive visual-technique selection and continuous intensity control in a real-time closed loop. As summarized in Table I, prior work either predicts multiple cognitive states without acting on them [6] or mitigates cybersickness alone with a single objective and fixed or predefined intensity. None jointly optimizes the four competing states, which is the gap AURORA fills.

Concern 2: The multi-task DL model was trained on the VRWalking dataset, while the RL agent was trained in a domain-randomized simulation; however, the RL reward function directly relies on the predictive outputs of this DL model. Nevertheless, the paper neither evaluates how the prediction errors of this DL model are amplified or propagated during RL training, nor provides an analysis of the RL agent's online prediction errors within a real-world user environment.

Response: We thank the reviewer for this comment. The predictions are converted into the reward through Equations 1 and 2, whose penalty terms prevent a single noisy prediction from driving a large policy update. To avoid optimizing an unreliable proxy, we evaluated the model across all four states, where it reaches 93%, 88%, 94%, and 90% accuracy for CS, CPL, CML, and WM with strong per-class F1 scores (Table II). We then measure the effect of prediction error on training directly through a robustness analysis in Section V that injects controlled noise into the predictions used for the reward and reports the change in the learned policy, whose outcomes remain stable under bounded noise will add noise range and resulting change in CS, CPL, CML, and WM], showing that prediction errors do not amplify during training. The policy is further validated independently in the user study, where the states are measured through FMS, NASA-TLX, and the digit span test rather than the model, and AURORA reduced cybersickness by 75.8% while lowering cognitive load and improving working memory, confirming the reward was aligned with real user states. Since the deployed policy is fixed and fine-tuned only offline, online prediction errors are not fed back into it, and a direct comparison of online predictions against user-reported states is left for future work.

Concern 3: The description of the HIL personalization process is vague and lacks reproducibility. Furthermore, critical details, such as the offline fine-tuning procedure, data volume, and number of fine-tuning rounds, are missing, rendering the method impossible to reproduce or to evaluate for its effectiveness.

Response: We thank the reviewer for this comment. The overall personalization mechanism, how the in-session and post-session feedback are combined into the user-specific preference vector $\mathbf{z}$ and used to re-weight the reward α_k and constrain the per-technique intensity ranges before offline PPO fine-tuning, is described in Section III (E) and in our response to Reviewer-1 (Concern 3). To make the procedure reproducible, we have expanded Section III (E) with the missing quantitative details. For each user, the in-session feedback consists of periodic self-reports of cognitive state collected at fixed intervals during immersion together with the concurrent MTL predictions of the remaining targeted states, and the post-session feedback consists of one structured preference questionnaire per session; these are aggregated into $\mathbf{z}$. The personalized policy is obtained by warm-starting from the general AURORA policy and continuing PPO under the $\mathbf{z}$-adjusted reward and intensity constraints, using the same PPO hyperparameters reported in Table II but for a reduced fine-tuning budget rather than full retraining. We now report the per-user feedback volume, the offline fine-tuning step budget, and the number of fine-tuning rounds performed between sessions, so that the personalization procedure can be reproduced and its effectiveness assessed. The quantitative benefit of this procedure is reported by the direct AURORA versus AURORA-HIL comparison in Table **??**.

Concern 4: The sequence was fixed as Baseline → Static → AURORA → AURORA-HIL. No Latin square or randomization procedures were employed, resulting in significant order and fatigue effects. Furthermore, participants were aware of the sequence of conditions, which may have introduced expectation bias.

Response: We thank the reviewer for this comment. This concern about a fixed sequence and the absence of Latin-square randomization is addressed in detail in our response to Reviewer-1 (Concern 1). We have revised Section VI accordingly.

Concern 5: Table VI employs the Wilcoxon signed-rank test; however, this test requires that paired data originate from the same distribution and exhibit symmetry—a condition that was not verified. Furthermore, the coefficient r is interpreted as the "rank-biserial correlation," yet the table fails to provide the corresponding Z-values. Moreover, some r values approach 1 or exceed it, which is theoretically impossible outcomes, suggesting errors in either the calculation or the reporting of the results.

Response: We thank the reviewer for these points on Table **??**. We have verified the symmetry of the paired differences for each comparison by inspecting their distributions, which were approximately symmetric and consistent with the use of the signed-rank test, and we report this in Section VI. We have also added the Z statistic to the table, and since the rank-biserial r is obtained from W and the sample size rather than from Z, the reported effect sizes are unchanged and Z is included for completeness. Additionally, none of the reported r exceeds one. The largest magnitude is 1.00, at DFOV vs. DGB for CPL, where $W = 0$. The rank-biserial correlation is bounded on $[-1, 1]$, and 1.00 is the valid endpoint reached when all paired differences share one sign, which is exactly the case at $W = 0$. Each r follows $r = 1 - 2W/(n(n+1)/2)$ from its own W and effective sample size, so the values near one reflect large and consistent effects rather than a calculation or reporting error. We have added this definition of r to Section VI so that the table can be read directly.

Concern 6: The comparative methods are limited to static techniques and a non-adaptive baseline, lacking comparisons against other adaptive approaches. Consequently, it cannot be demonstrated that AURORA's multi-objective RL strategy significantly outperforms simpler adaptive strategies.

Response: We thank the reviewer for this comment. This concern about the absence of comparisons against other adaptive approaches is addressed in detail in our response to Reviewer-1 (Concern 2).

Concern 7: RL training is conducted within a domain-randomized simulation; however, the user state transition model within this simulation is unknown and has not been fitted using real user data. Consequently, whether the resulting policy can effectively operate on real users remains, in essence, unverified.

Response: We thank the reviewer for this comment. The transition dynamics are not fitted to a single user model because real per-user dynamics are unknown and differ widely across individuals. We therefore train the policy in domain-randomized VR maze environments that vary layout, motion dynamics, visual flow, and walking speed, so it learns to generalize across a broad range of user responses rather than overfit one assumed model. The reward that guides this training is grounded in real user data, as the MTL model that produces it is trained on eye- and head-tracking signals with cognitive-state labels collected from real users during VR navigation. More importantly, the trained policy is evaluated on real users rather than only in simulation. In the user study with 47 participants on a consumer headset, the targeted states are measured with the FMS, NASA-TLX, and the digit span test rather than the model, and AURORA reduced cybersickness by 75.8% while lowering cognitive load and improving working memory, with the HIL variant further personalized from each user's feedback. This directly establishes that the policy operates effectively on real users, which is the validation that any fitted user simulator would itself ultimately require. We have made the rationale for domain randomization explicit in Section III (C) and now reference the user study (Section VI) as the real-user validation of the learned policy.

Minor Concern 8: The terms "cognitive mental load" and "CML" are used interchangeably within the main text; however, in Tables I and III, "CML" is inconsistently mixed with "cognitive mental load." Furthermore, Fig. 2 introduces "r_CML" and "r_CPL," yet these terms are not explicitly defined in the caption.

Response: We thank the reviewer for this comment. We have standardized the terminology so that CML is used consistently throughout the text and in Tables I and III, and we have defined r_{CML} and r_{CPL} in the caption of Fig. 2 of the revised manuscript.

Minor Concern 9: The units or normalization method for I_t are not explained in the reward definition.

Response: We thank the reviewer for this comment. We have clarified in the reward definition (Equation 2) that I_t denotes the normalized intensity of the active visual technique, and we describe its normalization and parameter ranges in Section III (D) of the revised manuscript.

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